



VEHICLE DYNAMICS HANDBOOK

BOOK FIVE

FROM ZERO TO EXPERT

First numbers • Simulation • Derivations • A worked car • Software • Mastery

TEAM BLITZ | MIST

The expert-level companion to Books One–Four

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BOOK FIVE

| FORMULA STUDENT VEHICLE DYNAMICS

From Zero to Expert

The expert-level companion — first numbers, simulation, derivations, a worked car, software, the deeper subsystems and theory.

About this book

Books One through Four take a Formula Student vehicle-dynamics package from the physics of the tyre to a built, validated, competition-ready car. They are deliberately practical, and in being practical they lean on numbers, simulations and tools as though those things were already available. This book fills the gap between the first four and true mastery: it explains how to produce the inputs the earlier books assume, how to simulate and reason about them rigorously, how the governing equations are actually derived, and how the deeper subsystems and theory fit together. It is the book to read second time through — once the vocabulary of Books One and Two is familiar — and the one to return to when a design decision needs more than a rule of thumb.

Eleven chapters, grouped loosely: getting started with no data (Chapter 1), simulating and reasoning (Chapters 2–4), executing in software (Chapter 5), the subsystems and the human that the first books touched only lightly (Chapters 6–7), the discipline of testing and proving (Chapters 8–10), and the handling theory that underlies all of it (Chapter 11).

How this book refers to the others

References such as “Book Two, Part E” or “Phase 3” point to the matching chapter of the main handbook. In the combined edition these become live links; read as a standalone supplement, they are simply signposts to where the groundwork was laid.

Chapter 1 — From zero: getting the first numbers

Every simulation, load case and target in the earlier books needs numbers that describe the car — its mass, where that mass sits, how the tyres behave. At the start of a project none of those numbers exist, because the car that would produce them has not been designed. This is the central paradox of first-year design: the analysis needs the car, and the car needs the analysis. The way out is not to wait for certainty but to start with disciplined estimates, label them as estimates, and tighten them in a loop. This chapter is how those first estimates are made and where they come from.

The convergence spiral

Design does not proceed in a straight line from inputs to outputs; it spirals inward. A first pass uses rough assumptions to produce a first design, that design yields better estimates of the same quantities, and the pass is repeated with the improved numbers. Each loop changes the outputs less than the last, and the process is “converged enough” when the quantities that drive decisions — lap time, balance, peak loads — move by less than a few percent between loops.

estimate v0 → design → measure / refine → estimate v1 → design → ...

where Convergence is reached when the decision-driving outputs change by less than a chosen tolerance (e.g. <3%) between successive loops. Record which version of each number a decision was made on.

The practical consequence is that the first numbers do not need to be right — they need to exist, be written down with their source and confidence, and be revisited. A blank cell blocks the whole team; a clearly-labelled estimate lets every sub-team start in parallel.

Mass and weight distribution from a mass budget

Before a car can be weighed it can be estimated, component by component, from a mass budget: a list of every significant mass, its value, and its position. Early values come from CAD, supplier data sheets, a previous car, or benchmark ranges; they are refined to measured masses as parts are made. The budget doubles as the live mass tracker for the whole project (Chapter 10).

Table 1-1. A starting mass budget. Ranges are typical; replace each with your own CAD or measured value as it becomes known. The sum is your first total-mass estimate.

Component group	Typical FSAE combustion range	Early source
Engine + fluids	25–45 kg (single → four-cyl)	Manufacturer spec
Chassis (frame + mounts)	25–40 kg	CAD + tube mass
Driver (per rules, strapped)	65–85 kg	Rules dummy / real driver
Drivetrain (diff, chain, axles)	10–20 kg	CAD + supplier
Suspension (all four corners)	20–35 kg	CAD
Wheels + tyres (set)	15–25 kg	Supplier
Bodywork + aero	5–20 kg	CAD / estimate
Electrical + cooling + fuel	10–20 kg	CAD / supplier
Total with driver	≈ 230–320 kg	Sum of the above

With each component’s mass and its longitudinal position measured from a datum (say the front axle), the centre-of-gravity position follows directly from a weighted average:

$$\mathbf{x}_{cg} = \Sigma(m_i \cdot \mathbf{x}_i) / \Sigma m_i \quad \text{front mass fraction} = (L - \mathbf{x}_{cg}) / L$$

where m_i , x_i = each component's mass and longitudinal distance from the front axle; L = wheelbase. The same sum in the vertical direction gives CoG height; in the lateral, the left/right split.

CoG height before a tilt test

CoG height is the single most influential number for load transfer, and the hardest to measure early because it needs the built car on a tilt table (Book Two, Part B). Until then it is estimated by the same weighted average applied to component heights:

$$h_{cg} = \Sigma(m_i \cdot h_i) / \Sigma m_i$$

where h_i = height of each component's own CoG above the ground. The low, heavy items (engine, driver hips, fuel) dominate; getting those three right gets h_{cg} close.

A useful discipline is to estimate the three or four heaviest items carefully from CAD and treat the rest as a lumped mass at a nominal height. The result is usually within a couple of centimetres of the eventual measured value — close enough to begin, and corrected the moment the car can be tilted.

Yaw inertia before the car can be swung

Yaw inertia governs how eagerly the car changes direction (Book Two, Part B) and is measured late, by swinging the finished car as a pendulum. Early on it is estimated from the same point-mass list used for the mass budget, by summing each mass times the square of its distance from the vertical axis through the CoG:

$$I_{zz} = \Sigma m_i \cdot (x_i^2 + y_i^2) \quad (= m \cdot k_z^2)$$

where x_i , y_i = component position relative to the CoG; k_z = radius of gyration. Items at the ends of the car (wheels, pedals, rear-mounted masses) dominate because of the square. A typical FSAE k_z is of order half the wheelbase.

This is why mass centralisation is a design goal and not only a mass-reduction one: moving a given mass toward the centre cuts its contribution to I_{zz} with the square of the distance. The CAD mass-properties tool computes this sum automatically once components are placed; the hand sum above is the sanity check.

Tyre behaviour before TTC data

Tyre data from the Tire Test Consortium (Book One, Part XI) is the foundation of the whole model, but access and fitting take time, and the first lap-sim runs are needed before that work is complete. Three stand-ins bridge the gap, in order of preference:

1. **Last year's fitted model** — if the team has run a tyre before, its fitted model is the best placeholder — the physics does not change between seasons.
2. **A published generic set** — published Magic-Formula coefficient sets exist for several common FSAE tyres (shared on the FSAE forum and in teaching material); load one into MFeval and use it until the real fit exists.
3. **Typical placeholder values** — when nothing else is available, seed the model with sensible placeholder values and flag every downstream number as provisional.

Table 1-2. A placeholder tyre to unblock the first simulations. Every value here is provisional and must be replaced by a fitted, scaled model before any decision is frozen.

Quantity	Placeholder to start with	Replace with
Peak lateral μ (track-scaled)	≈ 1.4 – 1.6	Scaled TTC fit at operating load
Cornering stiffness $C\alpha$	≈ 800 – 1400 N/deg at operating load	TTC fit slope (B·C·D)

Quantity	Placeholder to start with	Replace with
Peak slip angle	$\approx 5\text{--}9^\circ$ (load-dependent)	TTC fit
Load sensitivity	$\approx 10\text{--}20\%$ μ drop per +1 kN	TTC fit
Operating pressure / camber	$\approx 69\text{--}83$ kPa / -1 to -3°	Pressure & camber sweeps

Where first-pass numbers come from

No team starts truly blank; a great deal of representative data is public. The habit of an expert is to anchor every early estimate to a source rather than to invent it:

- **A previous team car** — the single best source — real masses, real geometry, real behaviour from an almost identical car.
- **Public FS design reports** — many teams publish design reports with full parameter tables; read several and bracket your estimates by what competitive cars use.
- **The FSAE forum** — the tyre subforum and the general forum hold years of shared data, Magic-Formula fits and benchmark numbers.
- **Textbook typical-value tables** — the appendices of the standard references (Book Two's reading list) and OptimumG material carry typical-value tables for exactly these quantities.

The rule that makes estimates safe

An estimate is only dangerous when it is mistaken for a measurement. Keep one parameter file for the car (Chapter 10) in which every number carries a source and a confidence — measured, CAD, benchmarked, or guessed — and a date. The simulation then reports not just an answer but how much to trust it, and the team knows exactly which numbers to chase down first.

Chapter 2 — Lap simulation in depth

Book Three uses a lap simulation to turn targets into frozen numbers, but treats the simulation itself as a black box. An expert chooses the right fidelity for the question, understands what each kind of model can and cannot say, and converts its output into competition points rather than seconds alone. This chapter opens the box.

The fidelity ladder

Lap simulations exist at increasing levels of detail, each answering a different question at a different cost. Using a model more complex than the question requires wastes time; using one too simple gives confident wrong answers.

Table 2-1. The lap-sim fidelity ladder. Most FSAE concept work lives in the point-mass and QSS rows; transient MBD is a late-stage refinement.

Model	What it represents	Answers well	Cannot tell you
Point-mass (GGV)	The car as one mass bounded by a g-g-v envelope	Lap time, event sensitivities, concept trade studies	Balance, yaw response, which axle limits
Bicycle	Two axles, yaw freedom, cornering stiffnesses	Understeer/oversteer, steady balance trends	Combined-slip limit behaviour in detail
Quasi-steady-state (QSS)	Full tyre model solved at each point; MMM per state	Limit balance, grip ceiling, setup direction	True transients (turn-in, mid-corner upsets)
Transient (full MBD)	Masses, compliances, dampers in time domain	Response, platform, kerb behaviour	— (most expensive to build and run)

The GGV envelope

At the heart of a point-mass sim is the GGV diagram — the surface of accelerations the car can reach as a function of speed. For every speed it defines the most the car can do in three directions, and the simulated driver then takes the car around the track always riding the edge of this envelope.

- **Maximum lateral (cornering)** — the tyres' lateral grip plus any aerodynamic download at that speed, divided by mass.
- **Maximum longitudinal accel** — whichever is smaller of what the tyres can transmit and what the engine can deliver through the gearing at that speed.
- **Maximum braking** — the tyres' longitudinal grip plus aero drag and download; usually the largest of the three.

$$a_{y,max}(V) \approx \mu_y \cdot (1 + F_{down}(V)/W) \text{ combined: } (a_x/a_{x,max})^2 + (a_y/a_{y,max})^2 \leq 1$$

where $F_{down}(V)$ = downforce at speed V ($\propto V^2$); W = weight. The combined-slip boundary is the friction ellipse from Book One, Part VII, applied at vehicle level.

Because download grows with the square of speed, the envelope widens at high speed for an aero car — which is the entire mechanism by which aerodynamics buys lap time, and why its benefit is track-dependent.

Modelling the events and spending effort on points

A lap time is only useful once it is converted into the competition's currency: points. The dynamic events reward different capabilities, so the same kilogram or degree of grip is worth different amounts in each. Simulating each event and weighting by its points shows where design effort pays.

Table 2-2. Typical FS dynamic-event point weightings. Endurance dominates, so reliability and consistent pace outrank a peak-grip advantage that cannot last the distance. Confirm the current year's values.

Event	Rewards	Typical points (verify rulebook)
Acceleration (~75 m)	Traction, power, low mass	≈ 100
Skidpad	Steady lateral grip, balance	≈ 75
Autocross	Agility, transient response, grip	≈ 125
Endurance	Grip + reliability + consistency over ~22 km	≈ 275
Efficiency	Low fuel/energy use with pace	≈ 100

With each event simulated, a sensitivity study answers the question that actually matters — not “how many seconds” but “how many points.”

$$\Delta \text{points} / \Delta \mathbf{x} = \Sigma_{\text{events}} \left(\partial \text{points}_{\text{event}} / \partial \text{time} \right) \cdot \left(\partial \text{time}_{\text{event}} / \partial \mathbf{x} \right)$$

where $\mathbf{x}$ = a design variable (mass, CoG, grip, downforce). Summed over events and weighted by points, this ranks design levers by competition value, not by lap time in one event.

Points per kilogram, in practice

Running the full event set with mass swept by ± 10 kg typically shows mass costing points across every dynamic event at once — acceleration most sharply. Expressed as points per kilogram, it becomes a hard currency the whole team can trade against: a part that saves 1 kg for a known cost and effort can be compared directly with a grip or aero gain worth the same points. This is how a mature team decides what to spend its limited time and money on.

The simulated driver

Every lap sim contains a driver model, and its assumptions shape the result. The simplest rides the GGV edge perfectly; more detailed ones add reaction time, a finite rate of building and releasing grip, and an imperfect line. Two cautions follow: a perfect-driver sim overstates what a student driver will extract, so absolute lap times read optimistic; and because the same optimistic driver is applied to every configuration, the sim remains reliable for comparing options even when its absolute numbers are high. Use it for differences, trust measured data for absolutes.

Chapter 3 — The math behind the formulas

The earlier books state the governing equations and trust them. Building a tool, defending a design to a sharp judge, or diagnosing a result that looks wrong all require knowing where those equations come from. This chapter derives the handful that matter most, each from a simple balance of forces or moments.

Lateral load transfer, derived

Consider the car cornering at steady lateral acceleration. The inertial force acts through the centre of gravity, a height h above the ground; the tyres react it at the contact patches, a track width t apart. Taking moments about the line joining the outer contact patches, the vertical load that must shift from the inner to the outer tyres to balance that moment is:

$$\Delta Fz_{\text{total}} = (m \cdot a_y \cdot h_{\text{cg}}) / t$$

where m = mass; a_y = lateral acceleration; h_{cg} = CoG height; t = track. This is a property of mass and geometry alone — springs change how it arrives, not how much.

The same argument in the longitudinal plane, taking moments about an axle with wheelbase L in place of track, gives the fore-aft transfer under braking and acceleration, $\Delta Fz = m \cdot a_x \cdot h_{\text{cg}} / L$.

Splitting the transfer between the axles

The total is fixed, but how it divides front-to-rear is the balance knob (Book Two, Parts C–D). At each axle the transfer arrives by three paths — elastic through the springs and bars as the body rolls, geometric through the links at the roll centre, and directly from the unsprung mass:

$$\Delta Fz_f = (a_y/g) \cdot [(K\varphi_f / K\varphi_{\text{total}}) \cdot W_s \cdot H / t_f + W_{sf} \cdot z_{rf} / t_f + W_{uf} \cdot h_{uf} / t_f]$$

where $K\varphi$ = roll stiffness (front / total); W_s, W_{sf} = total and front sprung weight; H = sprung-CoG height above the roll axis; z_{rf} = front roll-centre height; t_f = front track; W_{uf}, h_{uf} = front unsprung weight and its CoG height. The first term (elastic) is the one you tune with springs and bars.

The fraction of total roll stiffness placed at the front therefore sets the front's share of load transfer — the Total Lateral Load Transfer Distribution — and, through tyre load sensitivity, the balance. This is the derivation behind the TLLTD knob the earlier books turn.

The understeer gradient, derived

For steady cornering, each axle must run a slip angle large enough for its tyres to generate the lateral force its share of the mass demands. That required slip angle is the axle's load divided by its cornering stiffness. The steer angle the driver must apply is the geometric (Ackermann) angle for the corner plus the difference between the front and rear slip angles:

$$\delta = 57.3 \cdot L / R + K \cdot a_y \text{ with } K = W_f / C_{\alpha,f} - W_r / C_{\alpha,r}$$

where δ = steer angle (deg); L = wheelbase; R = corner radius; a_y in g; W_f, W_r = axle loads; C_{α} = axle cornering stiffness. $K > 0$ understeer, $K < 0$ oversteer, $K = 0$ neutral.

Two speeds fall straight out of this. The characteristic speed, where an understeering car needs twice its Ackermann steer, and the critical speed, where an oversteering car becomes unstable, both follow from K and the wheelbase — which is why a positive (understeering) K is designed in for stability.

Geometry from hardpoints

A kinematics tool computes camber, roll centre and bump steer from the suspension pickup coordinates; understanding the construction lets an expert read the tool's output and, if needed, build their own. The constructions all begin from the instant centre — the point about which the wheel instantaneously rotates, found by projecting the two control-arm axes until they cross in the front view.

- **Roll centre** — draw a line from each tyre's contact patch to its instant centre; where the two lines cross on the car centreline is the roll-centre height.
- **Camber gain** — the rate of camber change with wheel travel, set by the front-view swing-arm length (contact patch to instant centre); a short swing arm gives high gain.
- **Bump steer** — the toe change with wheel travel, found by matching the tie-rod's arc to the suspension's arc; designed to near zero.

camber gain $\approx - (\text{wheel travel}) / (\text{FVSA length})$ (small-angle, per unit bump)

where FVSA = front-view swing-arm length from contact patch to instant centre. Short FVSA $\rightarrow$ large gain, but the instant centre — and hence the roll centre — then moves a lot through travel (the migration trade of Book Two, Part E).

Building a Milliken Moment Method diagram

The Milliken Moment Method maps the car's balance across its whole operating range on one chart, and is built entirely from the tyre model and the mass data assembled above. The construction is a double sweep:

1. **Choose the grid** — sweep the body slip angle (vehicle sideslip) and the steer angle across their ranges.
2. **Per state** — for each combination, compute every tyre's slip angle from the geometry and motion, and read its lateral force from the tyre model at its current vertical load.
3. **Reduce** — sum the four tyre forces for total lateral acceleration, and sum force times longitudinal distance for the yaw moment about the CoG.
4. **Plot** — plot yaw moment against lateral acceleration. The envelope's rightmost point is the grip limit; where the curve crosses zero yaw moment is a trimmed (balanced) state; the slopes give stability and control authority.

Why the MMM is worth the effort

A lap sim says how fast; the moment method says why, and whether the car is controllable at the limit. It reveals, before any metal is cut, whether the car is stable, how much control the driver has left at peak grip, and which way the balance moves as the car is pushed — all from the same tyre and mass numbers already in hand. Bringing one to a design review signals a team that understands its car, not just its lap time.

Chapter 4 — A worked example, end to end

The preceding chapters, and Books One through Three, are easier to hold together when seen applied to a single car. What follows is one illustrative pass for a hypothetical autocross-focused combustion car, carried from targets to a first lap. Every number is illustrative and rounded; the purpose is the chain of reasoning, not the values.

Step 1 — Targets and first assumptions

Table 4-1. Starting point for the worked car. All provisional, all to be refined in the spiral.

Quantity	First assumption	Source (Chapter 1)
Total mass with driver	300 kg	Mass budget
Weight distribution	45% front	Mass budget
CoG height	290 mm	Weighted component heights
Wheelbase / track	1550 mm / 1200 mm	Rules + packaging
Tyre peak μ (scaled)	1.50	Placeholder (Table 1-2)
Target lateral grip	≈ 1.5 g	Tyre μ , no aero

Step 2 — Grip and a sanity check

With no aerodynamic download, steady lateral acceleration is approximately the tyre friction coefficient, so the skidpad grip target is about 1.5 g. On a 15.25 m skidpad that corresponds to a speed of roughly 46 km/h — a believable figure for the event, confirming the grip assumption is not absurd before any detailed work begins.

$$a_y \approx \mu_y = 1.5 \text{ g} \quad V_{\text{skidpad}} = \sqrt{(a_y \cdot R)} = \sqrt{(1.5 \cdot 9.81 \cdot 7.625)} \approx 10.6 \text{ m/s} \approx 38 \text{ km/h}$$

where R = skidpad path radius (≈ 7.6 m to the car's centreline). A plausible speed is the first check that the tyre and mass assumptions are self-consistent.

Step 3 — Springs from ride frequency

Choosing ride frequencies of 2.8 Hz front and 3.0 Hz rear (the non-aero band of Book Two, Part F), and a sprung corner mass of about 65 kg at the front, the required wheel rate and, through an assumed motion ratio of 0.7, the coil rate follow directly.

$$K_{\text{wheel}} = (2\pi f)^2 \cdot m = (2\pi \cdot 2.8)^2 \cdot 65 \approx 20100 \text{ N/m} \approx 20 \text{ N/mm} \quad K_{\text{spring}} = K_{\text{wheel}} / MR^2 = 20 / 0.49 \approx 41 \text{ N/mm}$$

where f = ride frequency; m = sprung corner mass; MR = motion ratio. The nearest available spring (say 40 N/mm) is chosen and the frequency re-checked — one turn of the spiral.

Step 4 — Roll and a load case

Targeting a roll gradient of 1.5 deg/g sizes the total roll stiffness, and the anti-roll bars make up whatever the springs alone do not provide, split front-to-rear to place the balance just forward of neutral. The lateral load transfer at the 1.5 g limit then sizes the worst-case tyre load that the upright and bearing must carry.

$$\Delta Fz_{total} = m \cdot a_y \cdot h / t = 300 \cdot (1.5 \cdot 9.81) \cdot 0.29 / 1.2 \approx 1070 \text{ N outer}$$

$$\text{front } Fz \approx \text{static} + \text{its share} \approx 660 + 480 \approx 1140 \text{ N}$$

where The outer-front vertical load (static plus transferred share) is the number the Chapter-on-load-cases work and the FEA (Book Three, Phase 4) are built around.

Step 5 — Close the loop

These first outputs feed back: the measured spring sets the true ride frequency, the built rocker sets the true motion ratio, the corner-weighted car sets the true weight distribution, and the tilt test sets the true CoG height. Each returns to the start of the chain and the pass is repeated. After two or three loops the numbers settle, the hardpoints can be frozen, and the car that began as a column of assumptions has become a defensible design — which is exactly the journey Book Three’s phases formalise.

Reading the worked example

Notice that no step required a number that could not be estimated up front, and that every estimate had a defined way to be replaced by a measurement. That is the whole method of Chapter 1 in miniature: start with sourced estimates, design, measure, and spiral inward until the car stops changing.

Chapter 5 — Software workflows

Book Three names the tools a vehicle-dynamics team uses but assumes someone already knows how to drive them. This chapter gives the workflow for each — the sequence of actions that produces the output the design phases ask for. Menu labels differ between versions, so what follows is the method, not the exact buttons; once the workflow is clear, the interface is easy to find.

Lap simulation (OptimumLap and similar)

1. **Build the vehicle** — define the vehicle: mass, weight distribution, a tyre grip coefficient or model, aerodynamic coefficients, engine torque curve, gearing and driveline efficiency.
2. **Build the track** — build or import the track from a centreline and corner radii (an autocross map or a GPS trace).
3. **Run** — run the simulation to get lap time and channel traces — speed, longitudinal and lateral g, the g-g usage.
4. **Sweep** — duplicate the vehicle, change one parameter, re-run, and compare. Repeating this across a range is the sensitivity study of Chapter 2.

Suspension kinematics (OptimumKinematics, Lotus, Adams/Car)

1. **Enter hardpoints** — enter the suspension pickup coordinates for one corner, plus wheel and tyre data.
2. **Run the sweeps** — run the three standard analyses: parallel wheel travel (both wheels bump together), opposite travel or roll, and a steering sweep.
3. **Read and iterate** — read the output curves — camber, toe, roll-centre height, half-track, caster — against wheel travel, and move pickups to hit the targets of Book Three, Phase 3.

In Adams/Car specifically the pattern is: build a double-wishbone suspension subsystem template from the hardpoints, assemble it into a suspension test rig, run the parallel-travel, opposite-travel and steering analyses, and request camber, toe, roll-centre and half-track as output measures. The free tools run the same three analyses with simpler interfaces.

Tyre model fitting and evaluation (MFeval, OptimumTire)

1. **Load** — load the raw TTC data for the chosen size, or an existing Magic-Formula coefficient file.
2. **Fit and scale** — fit the Magic Formula to the binned data (or use the supplied coefficients), then apply the friction scaling factor to make it track-representative.
3. **Evaluate** — evaluate lateral force against slip angle at each operating load and camber, and read off peak μ , cornering stiffness and peak slip angle for the model and the lap sim.

Structural FEA (ANSYS, SolidWorks Simulation)

1. **Mesh** — import the part and assign the material; mesh it, refining at fillets and load points.
2. **Load & constrain** — apply the Phase-4 load-case forces at the pickup points, with a dynamic magnification factor, and constrain the part as it is held in the car.
3. **Solve & check** — solve, then read peak stress against yield (for factor of safety) and peak deflection (for stiffness). Refine the mesh until the result stops changing — mesh independence.

Verify every tool against a hand calculation

A simulation is only trustworthy once it has agreed with a number computed by hand. Check a lap sim's skidpad speed against the grip formula; check an FEA stress against a simple beam or pin calculation; check a kinematics roll-centre against the geometric construction of Chapter 3. A tool that disagrees with a sound hand calc is wrong until the disagreement is understood — never the other way round.

Chapter 6 — Subsystem interactions

Three systems shape the car's behaviour as much as the suspension itself and were touched only lightly in the earlier books: the differential, the aerodynamic package, and the brakes. Each couples directly into balance and grip, and each is a vehicle-dynamics concern even when another sub-team owns the hardware.

The differential

The differential decides how drive torque splits between the rear wheels, and in doing so becomes a powerful balance tool at corner entry and exit. The common competition choice is a clutch-type (Salisbury) limited-slip differential, whose locking is set by ramp angles and preload.

- **Locking effect** — more locking makes both rear wheels turn together, which resists the car rotating — adding understeer on entry (off throttle) and stabilising the exit (on throttle) at the cost of some low-speed agility.
- **Ramps and preload** — separate ramp angles set how much the differential locks under acceleration and under deceleration, so entry and exit behaviour can be tuned independently; preload sets the locking at low torque.
- **The extremes** — an open differential maximises agility but spins the unloaded inside wheel out of slow corners; a spool (fully locked) maximises traction but fights every turn. The LSD is the tunable middle.

Because the differential changes balance on throttle and off, it is tuned alongside the dampers and bars (Book Two, Part M), not in isolation — an exit-oversteer problem is as likely to be solved at the differential as at the rear bar.

Aerodynamic–mechanical balance

When the car carries downforce, its balance has two contributors that blend with speed: the mechanical balance set by load transfer (constant with speed) and the aerodynamic balance set by where the downforce acts (growing with the square of speed). A car trimmed neutral at low speed can push or go loose at high speed as the aerodynamic balance takes over.

```
balance(V) = blend of TLLTD (mechanical, constant) and aero balance
% ( $\propto V^2$  share)
```

where Aero balance % = fraction of downforce on the front axle. As speed rises its weight in the blend grows, so the front/rear centre-of-pressure position relative to the CoG must be managed, not just the total download.

- **Ride-height sensitivity** — a ground-effect undertray's download changes sharply with ride height and rake, so the platform must be controlled (stiff enough springs, Book Two, Part F) or the aero balance wanders.
- **Feed it to the sim** — the aerodynamic map — lift, drag and balance as functions of ride height and rake — belongs in the lap sim, so the simulated balance is correct at every speed.

Brake bias

Braking shifts load forward, so the front tyres can carry more braking force than the rear; the brake system must deliver torque in the same proportion or one axle locks first. The front/rear torque split is set by the piston areas, rotor radii and pad friction, trimmed by the master-cylinder sizes and the pedal's balance bar.

```
T_brake = P_line · A_piston · μ_pad · r_eff ideal bias follows the
dynamic axle loads under braking
```

where P_{line} = line pressure; A_{piston} = caliper piston area; μ_{pad} = pad friction; r_{eff} = effective rotor radius. An adjustable balance bar lets the driver trim the front/rear line-pressure split to match the car and conditions.

The vehicle-dynamics interest is twofold: brake bias sets entry stability (too much rear locks the back and spins the car), and the forward load transfer it relies on is the same longitudinal transfer derived in Chapter 3 — so the brake calculation and the mass model must share one set of numbers.

Chapter 7 — The driver as a variable

The most powerful and least controlled element of the system sits in the cockpit. A setup change worth a tenth of a second is invisible next to a driver whose laps vary by a second, and the most precise balance in the world is wasted if the driver cannot feel or use it. Treating the driver as an engineered part of the loop is a mark of a mature team.

A shared feedback vocabulary

Driver feedback is only useful when it is specific and consistent, which means the team and the drivers must share a vocabulary. The essential structure is where in the corner the problem occurs and which way the car is unbalanced there.

Table 7-1. The standard debrief. Structured answers turn feel into the specific setup directions of Book Two, Part M.

Debrief question	What the answer isolates
Where does it happen — entry, mid, or exit?	Points to dampers/brakes (entry/exit) vs bars/springs (mid)
Does it push (understeer) or rotate (oversteer)?	The direction of the balance change
On or off throttle? On or off brakes?	Differential and brake-bias involvement
Consistent every lap, or changing?	Setup issue vs tyre temperature / fuel / driver
How confident do you feel at the limit?	Whether the car is usable, not just fast on paper

Consistency and the data check

A driver who is fast but inconsistent hides the effect of every setup change in lap-time scatter. The remedy is twofold: keep the same driver for back-to-back setup tests so the driver is a constant, and corroborate every subjective comment with the data traces — steering, brake pressure, throttle — which show what the driver actually did rather than what they felt. A reported “lose on entry” that the brake trace reveals as a late, hard stab is a driver input to coach, not a car balance to change.

Training as development

Seat time is development, not just practice. A driver who learns to place the car on its limit repeatably raises the whole team’s result and makes the data worth analysing. A short, structured debrief after every run — the questions above, written down against the setup sheet — builds both the driver and the record the team learns from.

The driver is part of the system

Grip, balance and response are only realised through the driver. A team that instruments and tunes the car but ignores the driver is optimising one half of the loop. The cheapest lap time most FSAE teams leave on the table is not in the car — it is in consistent, trained driving and a disciplined debrief.

Chapter 8 — Test method and statistics

Testing is how estimates become knowledge, but only if it is planned. An afternoon of changing things and “seeing what feels better” produces stories; a planned test produces numbers with known confidence. This chapter is the discipline that turns track time into trustworthy data.

Design of experiments

The default of changing one factor at a time is safe and clear, and right for most on-track tuning because it isolates cause. Its limits are that it is slow and misses interactions between factors. When several factors must be explored together, a structured matrix — a factorial or screening design — covers the combinations efficiently and reveals interactions, at the cost of more planning.

- **One factor at a time** — change one variable, hold the rest, attribute the result to that variable. Clear, slow, blind to interactions. The right default at the track.
- **Factorial / screening** — vary several factors across a planned grid to find main effects and interactions in fewer total runs; worth it when the factors are known to couple.
- **Replicate and randomise** — replicate key points and randomise run order so that drift — tyre wear, track temperature, fuel load — does not masquerade as an effect.

Uncertainty

Every measured and estimated number carries an error, and an answer without a sense of its uncertainty invites false confidence. The two questions to keep separate are sensitivity — how much the output moves when an input changes — and uncertainty — how well the input is known. A result that is sensitive to a poorly-known input is the one to chase down; a result insensitive to an input need not be measured precisely.

Sensitivity times uncertainty is where the risk lives

Rank every input by the product of how much the answer depends on it and how poorly it is known. The inputs at the top of that list — often CoG height, tyre scaling, and motion ratio early in a project — are where measurement effort buys the most confidence. This single ranking turns a vague “we should measure more” into a prioritised list.

Correlation and parameter identification

A model is validated not by looking right but by matching measurement. For every test, record measured against predicted and look at the residuals: a small, patternless scatter validates the model; a consistent offset or trend exposes an error to find. When the two disagree, the test data can often be used in reverse — to back-calculate the real value of a parameter that was assumed.

- **Identify C_α and balance** — from a constant-radius skidpad, the measured steer-vs-lateral-g slope yields the real understeer gradient, and with the axle loads, the effective cornering stiffnesses.
- **Identify roll stiffness** — from the measured roll angle versus lateral g, the real roll gradient — and hence the as-built roll stiffness and sprung-CoG arm — falls out.
- **Identify grip scaling** — from measured versus predicted skidpad grip, the true friction scaling factor that replaces the placeholder of Chapter 1.

This closing of the loop — measure what was assumed, feed it back into the model — is what makes a simulation trustworthy enough to design next year’s car from, and is the deepest meaning of the correlation loop named throughout Book Two.

Chapter 9 — Durability and fatigue

Endurance is roughly twenty-two kilometres of repeated load cycles, and a part that is strong enough to survive one corner can still fail from the millions of smaller cycles over a stint. Designing for the distance, not just the peak, is what separates a car that finishes from one that is fast until it breaks.

Static strength versus fatigue

A static factor of safety guards against a part yielding under the worst single load. Fatigue is different: a part cycled below its yield strength can still crack and fail if cycled enough times, because damage accumulates at stress concentrations. The governing picture is the stress-life (S-N) curve, which relates the stress amplitude of a cycle to the number of cycles the material survives.

- **Endurance limit** — for steels there is an endurance limit — a stress below which the part survives effectively forever. Designing the cyclic stress below it gives infinite life, the safe approach for suspension steelwork.
- **Aluminium** — aluminium has no true endurance limit; its strength keeps falling with cycles, so aluminium parts are designed to a defined life with margin, and inspected.
- **Stress concentrations** — cracks start at stress raisers — sharp fillets, weld toes, holes. Generous radii and good weld profiles raise fatigue life far more than adding bulk material.

Welds deserve particular care: the heat-affected zone and the weld toe are classic fatigue origins, which is why Book Four's fabrication rules keep welds away from peak-stress points and insist on clean profiles.

Bearing life

Wheel bearings are rated by the load they carry and are sized to a chosen life. The standard rating relates the dynamic load capacity and the applied load to the number of revolutions ninety percent of bearings survive:

$$L_{10} = (C / P)^p \times 10 \text{ revolutions } p = 3 \text{ (ball)}, 10/3 \text{ (roller)}$$

where C = dynamic load rating (from the bearing data sheet); P = equivalent dynamic load combining radial and axial components; L10 = life 90% of bearings exceed. Convert revolutions to distance through the rolling radius, and compare against a season's running.

The equivalent load P combines the radial and axial (cornering) components through factors in the bearing catalogue — which is precisely why a single ball bearing that cannot take the axial corner load is the wrong choice, and a pair of angular-contact or tapered-roller bearings is right.

Design for inspection

Durability is also a maintenance property. Parts that can be inspected and replaced between sessions turn a potential retirement into a scheduled change. A critical fastener that can be seen and torque-checked, a rod end that can be examined for play, an upright that can be swapped for a spare — each converts the risk of a hidden failure into a routine check, which over an endurance event is worth as much as raw strength.

Chapter 10 — Data, mass budget and knowledge management

A Formula Student team rebuilds itself every year or two as members graduate, and the knowledge that is not written down leaves with them. The practices in this chapter are how a team stops starting from scratch each cycle — the organisational side of becoming expert, and the reason this handbook exists at all.

One source of truth: the parameter file

The surest way to produce an inconsistent car is to let the lap sim, the CAD, the FEA and the setup sheet each carry their own copy of the car's numbers. A single parameter file — one place that holds the canonical mass, CoG, track, tyre coefficients and the rest, each with its source, confidence and date — that every tool reads from keeps the whole project consistent and makes the convergence spiral of Chapter 1 auditable.

The living mass budget

The mass budget begun in Chapter 1 is not a one-time estimate but a living document, updated from estimate to CAD to measured value as each part is realised, with a target and a running total. It is simultaneously the mass estimate the sim needs, the budget the team is held to, and the early warning when the car is drifting heavy.

Version control

Scripts, simulation files and analysis deserve the same version control that software gets. Keeping them in a repository — with meaningful commit messages and clear file naming — means a result can always be reproduced, a change can be traced, and two members cannot silently overwrite each other's work. The cost is an afternoon of setup; the return is never losing last month's working configuration.

Rigorous concept selection

When several concepts compete — a tyre size, a damper, a material — the decision is made defensible by a weighted decision matrix rather than by preference. Each option is scored against criteria weighted by importance, and the scores are summed; the exercise forces the criteria to be named and the trade-offs to be explicit, which is exactly what a design judge probes.

1. **Define** — list the options and the criteria that matter (grip, mass, cost, data availability, manufacturability).
2. **Weight** — weight each criterion by its importance to the competition result.
3. **Score** — score each option against each criterion, multiply by the weights, and sum. The highest total is the defensible choice — and the reasoning is recorded.

The compounding return of writing it down

The reason top teams appear to “just know” their car is not talent but accumulation: years of parameter files, mass budgets, test logs and handbooks that each generation inherits and extends. Every number recorded with its source, every test logged, every decision captured in a matrix is a deposit the next team withdraws. This handbook is that principle applied to knowledge itself.

Chapter 11 — Advanced handling theory

The steady-state balance of Book Two explains the car at a constant cornering state; the car is also a dynamic system that responds to inputs over time, and understanding that response is the last step toward mastery. This chapter sketches the transient theory — enough to connect the earlier rules of thumb to the mathematics that underlies them.

The bicycle model

The simplest model that captures handling collapses the car to two axles on a centreline, with the front and rear cornering stiffnesses and the mass and yaw inertia as its parameters. Its two states — the lateral velocity and the yaw rate — evolve according to a pair of linear equations whose coefficients are built from those four quantities. From this compact model the whole of linear handling follows.

- **Understeer** — the balance of the front and rear cornering stiffnesses against the axle loads reproduces the understeer gradient of Chapter 3, now as a property of a dynamic system.
- **Characteristic and critical speed** — the understeering car has a characteristic speed at which its response is most benign; the oversteering car has a critical speed above which it is unstable. Both emerge directly from the model's coefficients.

Stability, damping and response

Because the car is a second-order system, it has a natural frequency and a damping ratio for its yaw motion, just as a spring-mass-damper does. These set how quickly the car responds to a steering input and whether it settles cleanly or overshoots — the quantities a step-steer test measures on track (Book Two, Part L).

- **Response time** — how fast the car builds yaw rate after a steering input; higher is sharper, set by cornering stiffness and yaw inertia.
- **Yaw damping** — whether the yaw motion settles smoothly or rings; too little and the car feels nervous, too much and it feels slow.
- **Frequency response** — how the yaw-rate response to steering changes as inputs get quicker; a flat response to high-frequency steering is a car that stays predictable when the driver is busy.

This is the theory beneath the mass-centralisation argument made all the way back in Book Two: a low yaw inertia raises the natural frequency and sharpens the response, so the same balance feels more responsive in a car that carries its mass near the centre. The rules of thumb and the mathematics meet here.

Where theory returns to the track

None of this replaces testing — it explains it. The step-steer response, the yaw-rate trace, the feeling of a car that is “darty” or “lazy” are the bicycle model's natural frequency and damping made physical. An engineer who can move between the two — reading a data trace as the behaviour of a second-order system, and predicting a setup change from the model — has closed the last gap between understanding and mastery. That is the destination this handbook was written to reach.